N021AV (S/N: W14031868922) — NVMe Media Errors Diagnostic Report

Immfly Ground Operations / IT Session: October 20, 2026

Field	Value
Machine	**N021AV** (S/N: W14031868922)
Affected component	Internal NVMe SSD (`/dev/nvme0n1`)
Initial symptom	Disk image backup to the external drive failed. `fsarchiver savefs` of `nvme0n1p2` reached 100% but reported 29 regular files with errors and exited with status 1, which aborted `save_server.py`.
Disk identification	FAILING — The kernel log shows repeated `critical medium error` READ failures on `nvme0n1` at many different sectors. SMART reports 676,835,408 Media and Data Integrity Errors.
Root cause	Physical media failure of the internal NVMe SSD. The external drive (`sda`) shows no I/O errors, so it is not the source of the errors.
Backup status	The image set is **not valid**. `backup-nvme0n1p1.fsa` and `backup-nvme0n1p2.fsa` were written in this session (the latter is incomplete); `p4` and `p5` are dated August 25 and were not regenerated because the script stopped at `p2`.
Recommended action	**Replace the NVMe SSD.** Do not use the generated image for restores. Rebuild the image from a healthy unit.

NARRATIVE SUMMARY

A full disk image of N021AV was being saved to the external drive with `python3 save_server.py save nvme0n1`, which archives each partition with `fsarchiver`. The archive of `nvme0n1p2` ran to 100% (138,797 regular files, 24,739 directories, 15,160 symlinks, 120 hardlinks and 17 specials processed), but 29 regular files could not be read and fsarchiver exited with status 1. The Python wrapper treated that exit status as a failure and stopped with `CalledProcessError`, so the remaining partitions were not saved.

The kernel log shows the cause: repeated `blk_update_request: critical medium error, dev nvme0n1` on READ operations, spread over many unrelated sectors (for example 1253088, 13836160, 17956728, 30540840, 34812920 and 38925608). The errors are in the disk itself, not in a single file or file system structure.

SMART data confirms it: 676,835,408 Media and Data Integrity Errors. The other SMART fields look healthy (Critical Warning 0x00, Available Spare 100%, Percentage Used 12%), so the drive does not flag itself as critical even though it is returning unreadable sectors. The external drive was checked in the kernel log and shows only normal attach messages, with no I/O errors.

In the backup folder, `backup-nvme0n1p1.fsa` (4.6 MB) and `backup-nvme0n1p2.fsa` (2.5 GB) carry today's date, while `p4` (16 GB) and `p5` (363 GB) are dated August 25. The folder therefore mixes files from different runs, and the `p2` archive from this disk is missing the 29 unreadable files.

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TROUBLESHOOTING STEPS PERFORMED

1. **Backup attempt:** `save_server.py save nvme0n1` failed after `fsarchiver -o savefs backup-nvme0n1p2.fsa /dev/nvme0n1p2 -Z1 -j4 -v` returned status 1 with 29 regular files in error.
2. **Kernel log check:** `dmesg` filtered for nvme and I/O errors showed repeated critical medium errors on READ at many sectors of `nvme0n1`.
3. **External drive check:** The kernel log for `sda` (1 TB) showed only normal attach and mount messages, with no I/O errors.
4. **SMART check:** `smartctl -a /dev/nvme0n1` reported 676,835,408 Media and Data Integrity Errors, with Critical Warning 0x00, Available Spare 100% and Percentage Used 12%.
5. **Backup folder check:** The folder contains `p1` and `p2` from this session and `p4` and `p5` from August 25. The image set is inconsistent and must not be used for restores.

CONCLUSION AND RECOMMENDATION

The internal NVMe SSD of N021AV has a physical media failure. File-level backup tools skip the unreadable files, so they cannot produce a faithful image of this disk. The same pattern was seen on N041AV.

Recommended action: Issue a maintenance work order to replace the NVMe SSD. Do not use the images created in this session for restores, and do not reuse the backup folder without confirming which run each file belongs to. If the contents of this disk are needed, a sector-level copy with `ddrescue` (which skips and logs unreadable sectors) is the appropriate tool, and it should be done before the disk degrades further.

APPENDIX: KEY COMMAND OUTPUT

```
[N021AV] fsarchiver result (nvme0n1p2)
Statistics for filesystem 0
* files successfully processed:....regfiles=138797, directories=24739, symlinks=15160, hardlinks=120, specials=17
* files with errors:...............regfiles=29, directories=0, symlinks=0, hardlinks=0, specials=0
subprocess.CalledProcessError: Command 'fsarchiver -o savefs backup-nvme0n1p2.fsa /dev/nvme0n1p2 -Z1 -j4 -v' returned
non-zero exit status 1.
# dmesg | grep -iE "nvme|blk_update_request|I/O error|medium" | tail (excerpt)
[ 1148.636991] blk_update_request: critical medium error, dev nvme0n1, sector 1253088 op 0x0:(READ)
[ 1152.921293] blk_update_request: critical medium error, dev nvme0n1, sector 38925608 op 0x0:(READ)
[ 1157.477071] blk_update_request: critical medium error, dev nvme0n1, sector 18038392 op 0x0:(READ)
[ 1159.385566] blk_update_request: critical medium error, dev nvme0n1, sector 13844144 op 0x0:(READ)
[ 1159.488695] blk_update_request: critical medium error, dev nvme0n1, sector 34812920 op 0x0:(READ)
[ 1159.614319] blk_update_request: critical medium error, dev nvme0n1, sector 30540840 op 0x0:(READ)
[ 1190.245658] blk_update_request: critical medium error, dev nvme0n1, sector 17956728 op 0x0:(READ)
[ 1192.177178] blk_update_request: critical medium error, dev nvme0n1, sector 13836160 op 0x0:(READ)
# external drive (sda) in the kernel log
[ 7.622449] sd 0:0:0:0: [sda] 1953525168 512-byte logical blocks: (1.00 TB/932 GiB)
[ 7.690897] sda: sda1 sda2
(no I/O errors reported for sda)
# smartctl -a /dev/nvme0n1 (excerpt)
Critical Warning: 0x00
```

```
Available Spare: 100%
Available Spare Threshold: 10%
Percentage Used: 12%
Media and Data Integrity Errors: 676,835,408
# ls -lh /mnt/usb/p100_restore_EFI | grep fsa
-rw-r--r-- 1 root root 4.6M Oct 20 15:52 backup-nvme0n1p1.fsa
-rw-r--r-- 1 root root 2.5G Oct 20 16:15 backup-nvme0n1p2.fsa
-rw-r--r-- 1 root root 16G Aug 25 13:45 backup-nvme0n1p4.fsa
-rw-r--r-- 1 root root 363G Aug 25 14:24 backup-nvme0n1p5.fsa
```